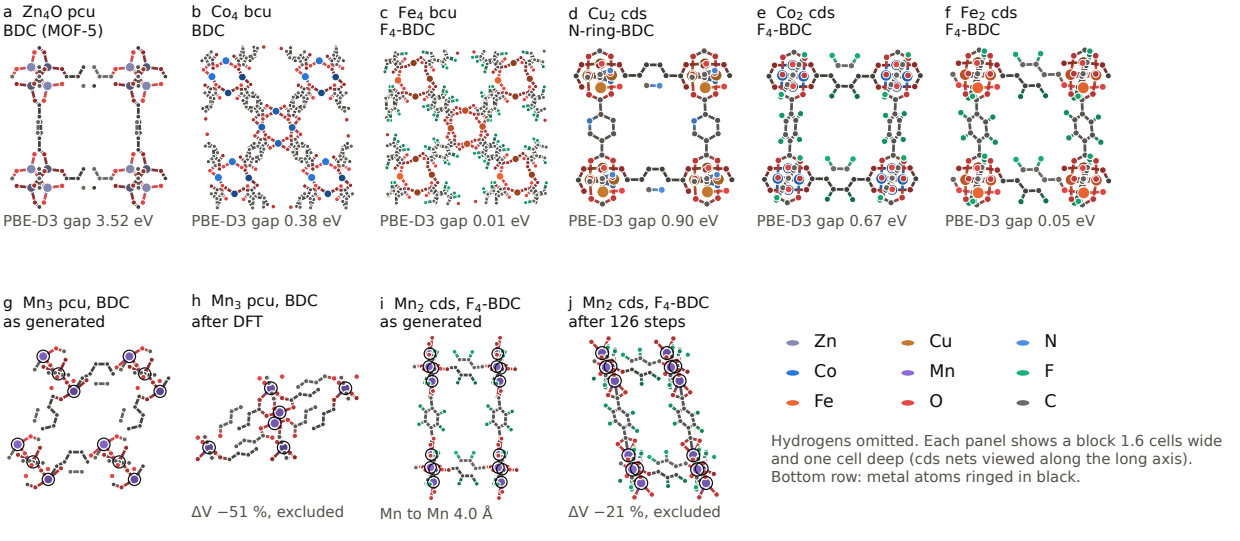

a Zn4O pcu
BDC (MOF-5)
PBE-D3 gap 3.52 eV
b Co4 bcu
BDC
PBE-D3 gap 0.38 eV
c Fe4 bcu
F4-BDC
PBE-D3 gap 0.01 eV
d Cu2 cds
N-ring-BDC
PBE-D3 gap 0.90 eV
e Co2 cds
F4-BDC
PBE-D3 gap 0.67 eV
f Fe2 cds
F4-BDC
PBE-D3 gap 0.05 eV
g Mn3 pcu, BDC
as generated
h Mn3 pcu, BDC
after DFT
ΔV −51 %, excluded
i Mn2 cds, F4-BDC
as generated
Mn to Mn 4.0 Å
j Mn2 cds, F4-BDC
after 126 steps
ΔV −21 %, excluded
Zn
Co
Fe
Cu
Mn
O
N
F
C
Hydrogens omitted. Each panel shows a block 1.6 cells wide and one cell deep (cds nets viewed along the long axis). Bottom row: metal atoms ringed in black.